



UFR Sciences et Technologies
Master2 E3A: Smart Aerospace & Autonomous Systems

Differential methods for noisy signal differentiation: Towards a new generation of model-free controllers

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OUTLINE

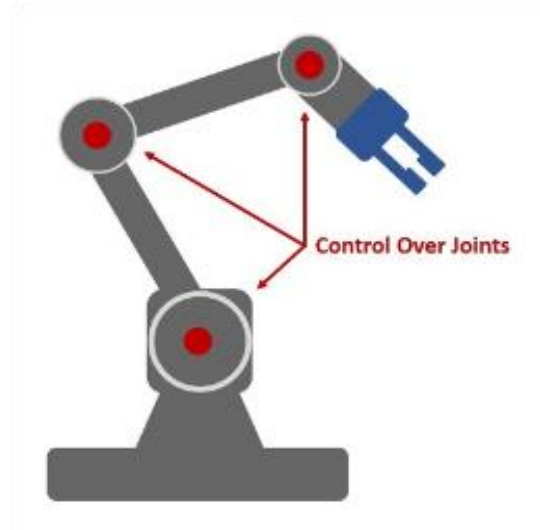
- Introduction
- Objective
- Literature Review
- Contribution
- Model-free control
- Simulation and results
- Conclusion and recommendation
- References

INTRODUCTION

What is Signal Differentiation?

- ✓ Computes the derivative of a signal,
- ✓ Fundamental operation in control systems, state estimation, and signal processing.

Importance of signal differentiation



<https://www.therobotreport.com>

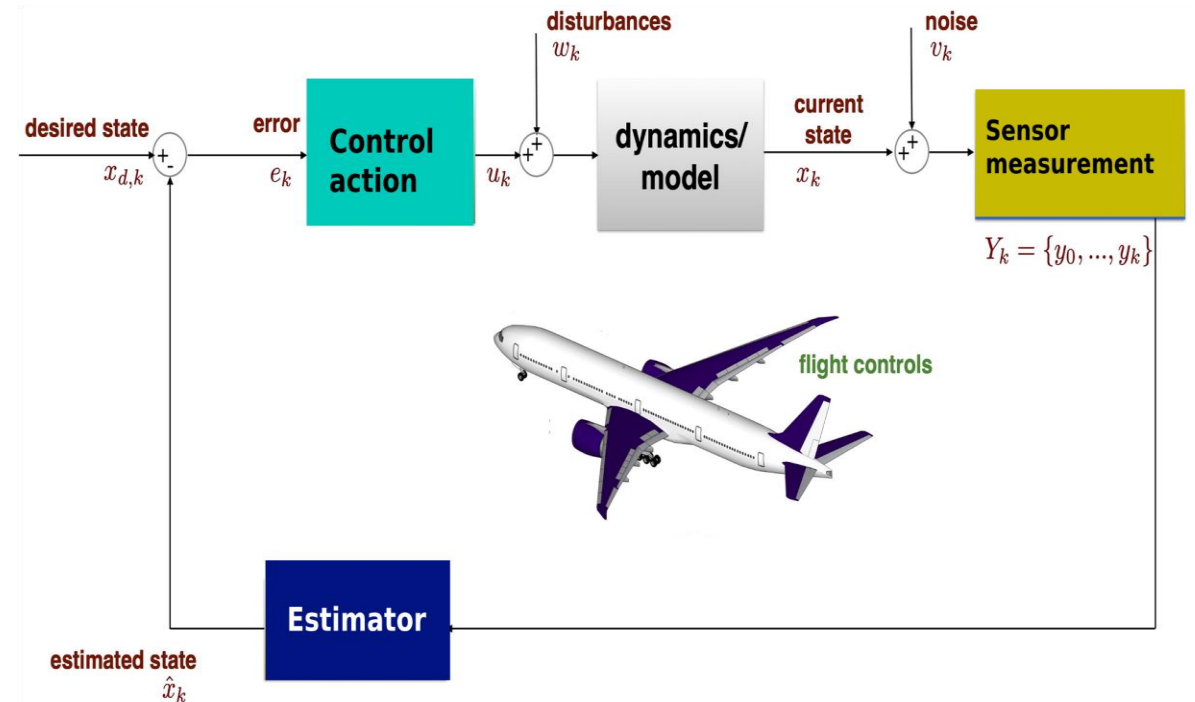


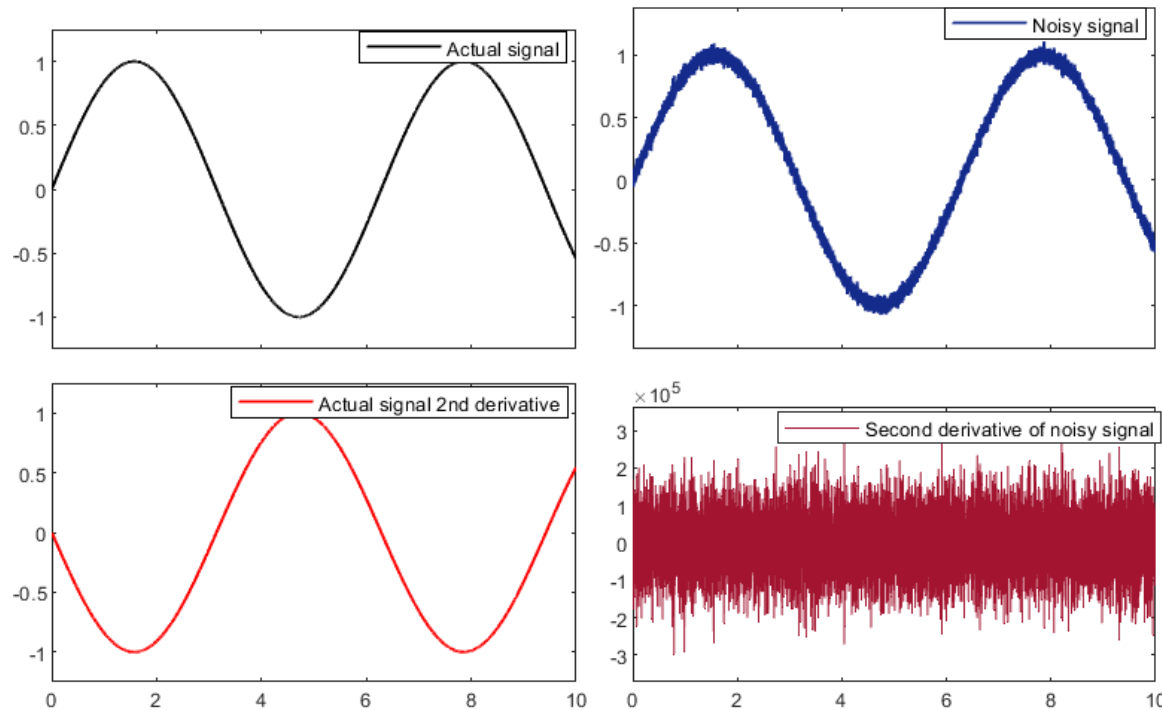
Fig: application area of signal differentiation [9]

The Challenges of Signal differentiation in noisy environment

- Noise Amplification
- Unstable Results

How to reduce the effect of noise?

- ✓ Low-Pass Filtering
- ✓ Smoothing a Noisy Signal
- ✓ Regularization Techniques
- ✓ Advanced Differentiation Methods
 - High-Gain Observers
 - Numerical differentiation with annihilators
 - Higher-Order Sliding Mode (HOSM) Differentiators



OBJECTIVE AND METHODOLOGY

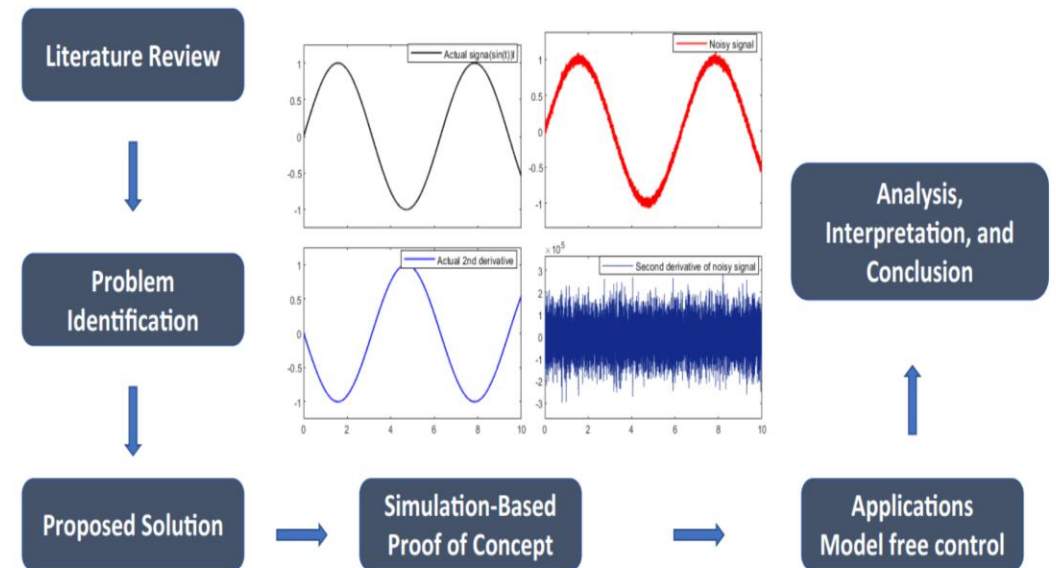
Objective

- To study, understand the algorithms and test them on Matlab for different signals and different noises.
- Apply the methods for model free control application

The algorithms

- ✓ High-gain observers for signal differentiation
- ✓ High-order sliding mode differentiators
- ✓ Algebraic algorithm (from Michel Fliess and Cédric Join)

Methodology



LITERATURE REVIEW

High-Gain Observer (Observable Canonical Form)

- ✓ Observer gain is determined using a **Lyapunov-like dynamic equation**
- ✓ Differentiation error is **bounded** and **tunable** via a single parameter
- ✓ Sensitive to noise due to high-gain terms appearing throughout the equations.
- ✓ Higher gains lead to faster convergence but also more pronounced peaking

Peaking Phenomenon

- ✓ High gains can lead to initial transient errors, known as **the peaking phenomenon**, which can impact accuracy and stability.

$$\begin{aligned}\dot{x}_i(t) &= x_{i+1}(t) + C_n^i \gamma^i (y(t) - x_1(t)), \quad i = 1, \dots, n-1, \\ \dot{x}_n(t) &= C_n^n \gamma^n (y(t) - x_1(t)).\end{aligned}$$

Controllable canonical (Brunovski) form

- ✓ Output to be differentiated appears as a control input
- ✓ The differentiation gain is determined using a Dual **Lyapunov-like dynamic equation**
- ✓ Less sensitive to noise because differentiation gain is smoother and only applied at the last state equation.
- ✓ More suitable for closed-loop control due to its controllable canonical structure.

$$\dot{x}_i(t) = x_{i+1}(t), \quad 1 \leq i \leq n-1,$$

$$\dot{x}_n(t) = -\gamma^n (x_1(t) - y(t)) - \sum_{i=1}^{n-1} C_n^i \gamma^i x_{n-i+1}(t),$$

Numerical Differentiation with Annihilators

- ✓ Uses algebraic differentiation techniques
- ✓ Based on annihilators to eliminate undesired terms
- ✓ Expressing derivative estimates as iterated integrals of the noisy signal over a short time interval
- ✓ Delay-Based Estimation for improved accuracy
- ✓ FIR Filter Implementation for Real-Time Numerical Differentiation

$$\begin{aligned}\tilde{x}^{(n)}(0; \kappa, \mu) &= \gamma_{\kappa, \mu, n} \int_0^1 \tau^{\kappa+n} (1 - \tau)^{\mu+n} x^{(n)}(T\tau) d\tau \\ &\quad + (-1)^n \frac{\gamma_{\kappa, \mu, n}}{T^n} \int_0^1 \frac{d^n}{d\tau^n} \{ \tau^{\kappa+n} (1 - \tau)^{\mu+n} \} \varpi(T\tau) d\tau\end{aligned}$$

Higher-Order Sliding Mode (HOSM) Differentiators

✓ Why we call it higher order? what is its importance?

✓ Homogeneous

✓ Finite-Time Convergence

✓ Handle bounded noise effectively based on Lipschitz boundary

$$\dot{z}_{-2} = u(t) - \gamma z_{-2},$$

✓ Chattering and reduction methods

✓ Extracting the equivalent control in (SMC) systems

$$\dot{z}_{-} = \mathbf{D}_k(z_{-}, z_{-2}, L, \lambda) - e_0 \gamma z_{-2}$$

$$\mathbf{D}_k(z_{-}, z_{-2}, L, \lambda) = dz_{i+1} = -\lambda_{k+1-i} L^{\frac{i+1}{k+1}} \left(|\text{error}|^{\frac{k-i}{k+1}} \text{sign}(\text{error}) \right) + z_{i+2}$$

Novel Differentiation Method

✓ Uses HOSM differentiation to estimate the equivalent control signal and its derivatives from a sliding-mode control input.

CONTRIBUTION

Proposed Methods : Modified High-Gain Observer Methods

$$\dot{x}_i(t) = x_{i+1}(t) + C_n^i (t/a + e)^i \gamma^i (y(t) - x_1(t)), \quad i = 1, \dots, n-1,$$

$$\dot{x}_n(t) = C_n^n (t/a + e)^n \gamma^n (y(t) - x_1(t)).$$

- t - time bound by $\tanh()$,
- $a > 0$ - is a correction factor which is a compromise between peak and accuracy
- e - epsilon to get non-zero gain at $t = 0$

Proposed Methods : Modified Brunovski Methods

$$\dot{x}_i(t) = x_{i+1}(t), \quad 1 \leq i \leq n-1,$$

$$\dot{x}_n(t) = -(\gamma(t/a) + e)^n (x_1(t) - y(t)) - \sum_{i=1}^{n-1} C_n^i (\gamma(t/a) + e)^i x_{n-i+1}(t),$$

- t - time bound by tanh(),
- a > 0 - is a correction factor which is a compromise between peak and accuracy
- e - epsilon to get non-zero gain at t = 0

✓ Eliminated the peak phenomenon and improve the noise rejection

Why these two modifications?

- ✓ Bounded at t = 0 & t = inf, peak eliminated, sensitivity to noise reduced, & comparable result with HOSM
-

MODEL FREE CONTROL (MFC)

What is model free mean?

- ✓ Designing a controller without knowing the exact dynamics of the plant (no complete mathematical model)
- ✓ Plant = known + unknown dynamics
- ✓ The controller will estimate the uncertain/unknown dynamics while keep tracking the known dynamics.

Why MFC? Complex, nonlinear, difficulty

Key features? Data-driven, adaptive & robust, no model

Example? PID, iPID, RL, FL, ADRC

How we integrate the noisy signal differentiation with MFC?

Application Area

- ✓ Aerospace
- ✓ Robotics
- ✓ Rocket and satellite control

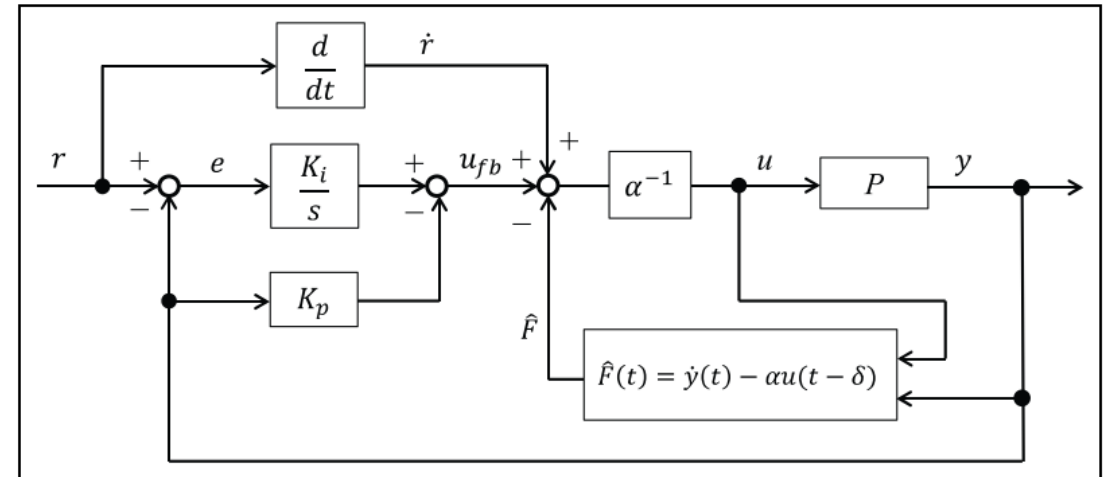
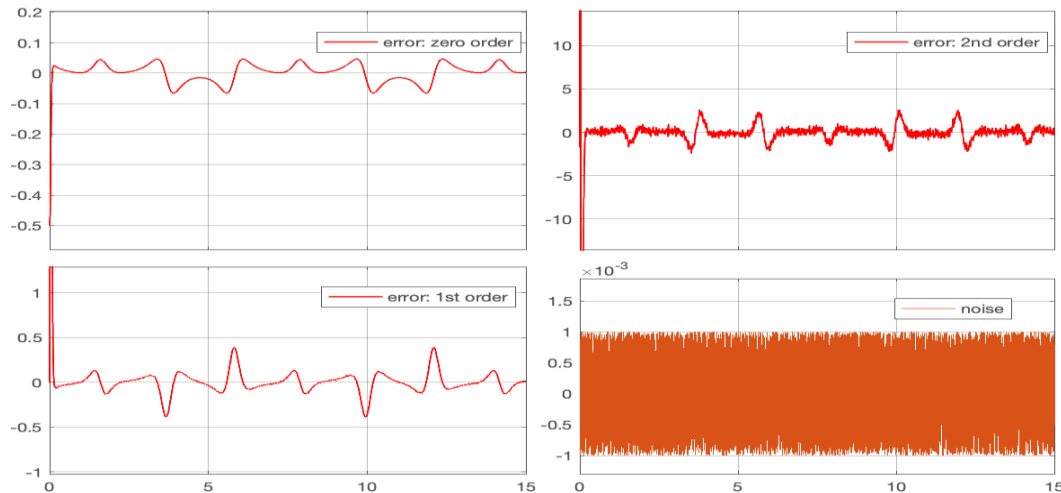


Fig: Model-free control (MFC) with an ultra-local model. [8]

SIMULATION RESULTS

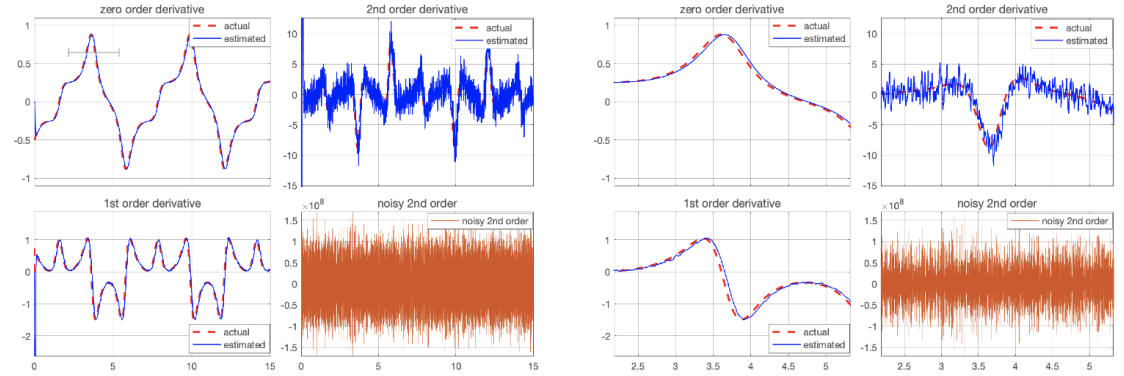
Brunovski Form

- Brunovski form results showed less peak and better computational speed than Observer based
- Noise reduction efficiency depends on the type of noise
- Accuracy is compromised compared to observer based

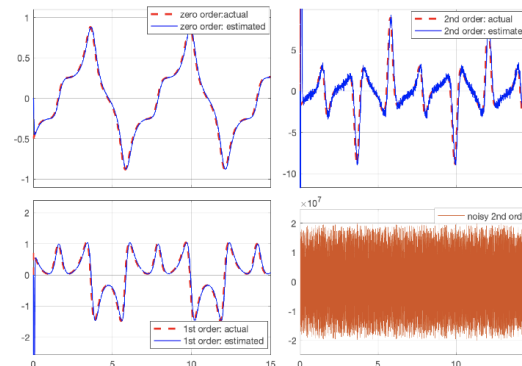


Error plot for Brunovski form

Estimated and actual derivative plots

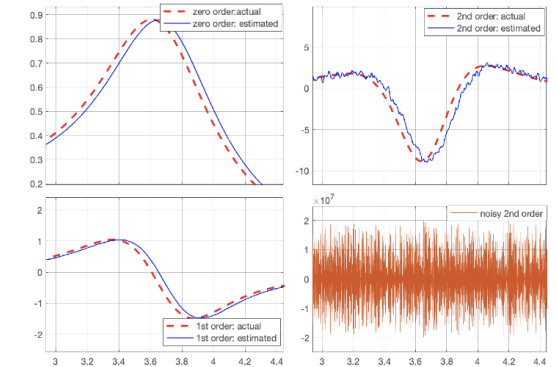


(a) White noise with $\sigma^2 = 1e-8$



(c) Uniform random noise of $\text{abs}(\text{mag}) \leq 1e-3$

(b) White noise with $\sigma^2 = 1e-8$ zoomed

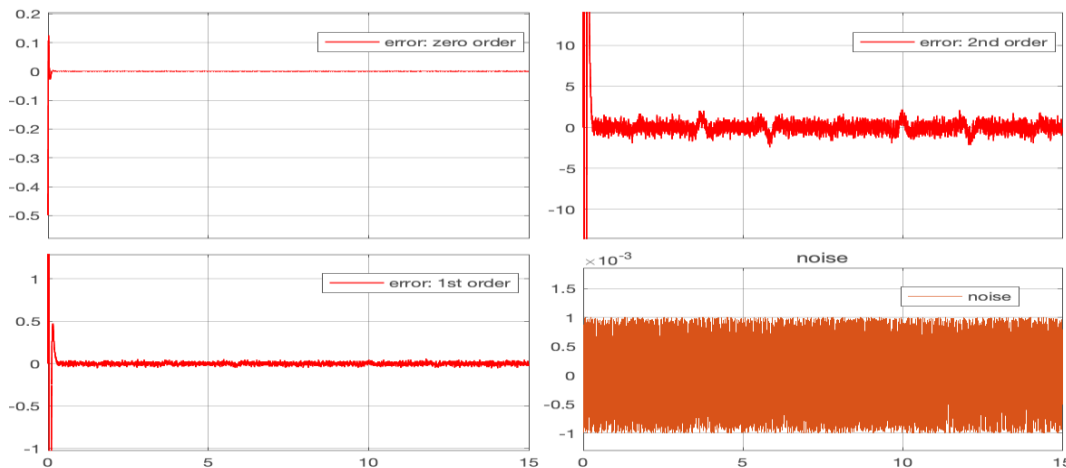


(d) Uniform random noise of $\text{abs}(\text{mag}) \leq 1e-3$ zoomed

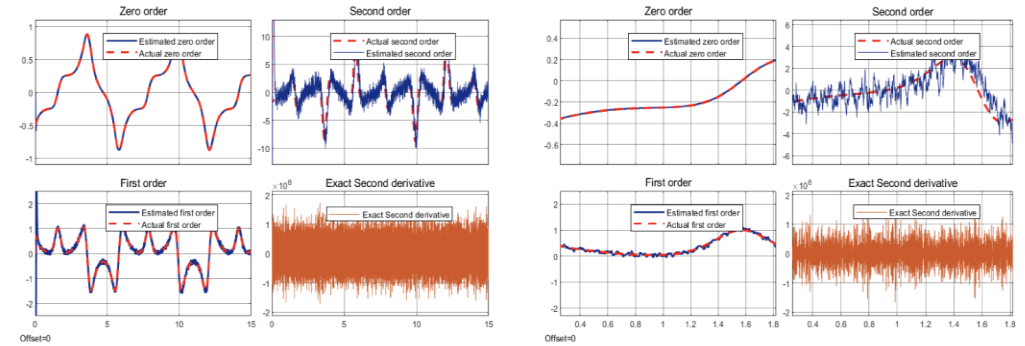
Figure 4.1: Ibrir Brunovski form high gain estimation for $y(t) = -\cos(t)/(2 + \sin(3t))$

HGO based differentiation

- results have high peak and less computational speed than previous one
- Noise reduction efficiency depends on the value of finite time d/ce ($1/\gamma$)
- Accuracy is better than Brunovski form if noise is zero

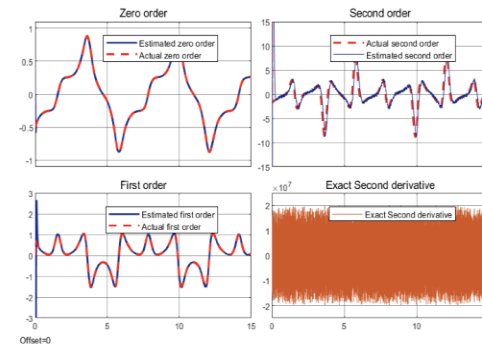


Error plot for the HGO uniform noise

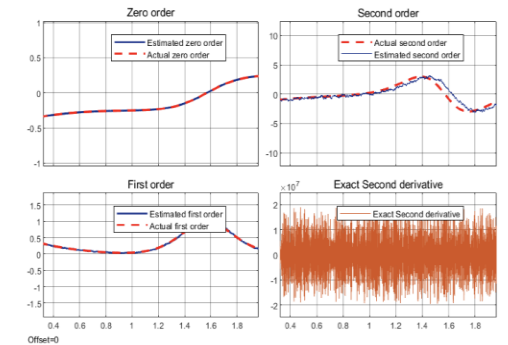


(a) White noise with $\sigma^2 = 1e - 8$

(b) White noise with $\sigma^2 = 1e - 8$ zoomed



(c) Uniform random noise of $\text{abs}(\text{mag}) \leq 1e-3$

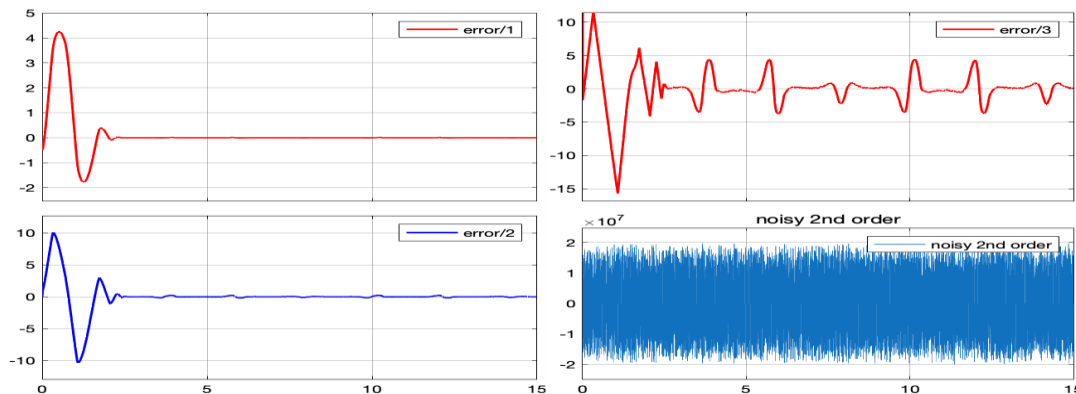


(d) Uniform random noise of $\text{abs}(\text{mag}) \leq 1e-3$ Zoomed

Figure 4.3: High-gain observer estimation for $y = -\cos(t)/(2 + \sin(3t))$

Sliding mode differentiators

- ✓ Insensitive to noise
- ✓ Robust for uncertainty
- ✓ Effectiveness depend on Lipschitz boundary
- ✓ Noise rejection efficiency better than HGO
- ✓ If there is no noise, perfect tracking



Error plot for different orders

Estimated and actual derivatives plot

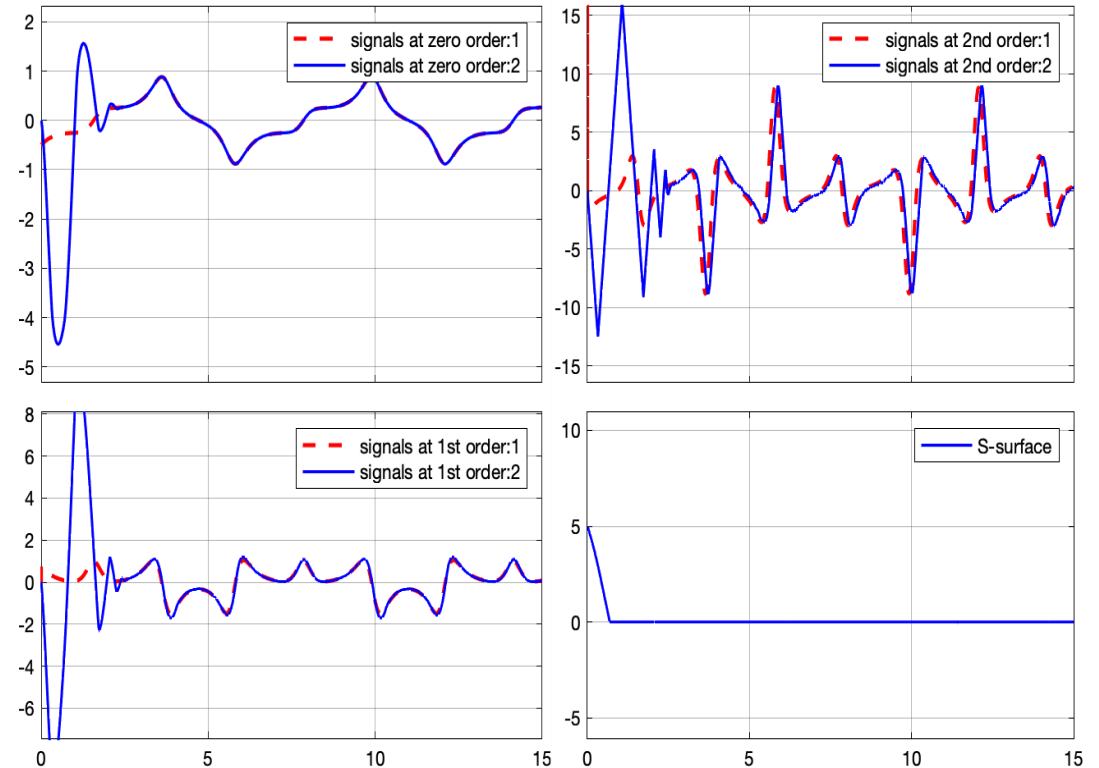


Fig: Actual (red) and estimated (blue) comparison

MODIFIED HGO

The modified version:

- ✓ Completely eliminated the peak phenomenon at the beginning
- ✓ Its noise rejection efficiency is better than both HGO and Brunovski form proposed by Ibrir[1]
- ✓ Less sensitive to noise

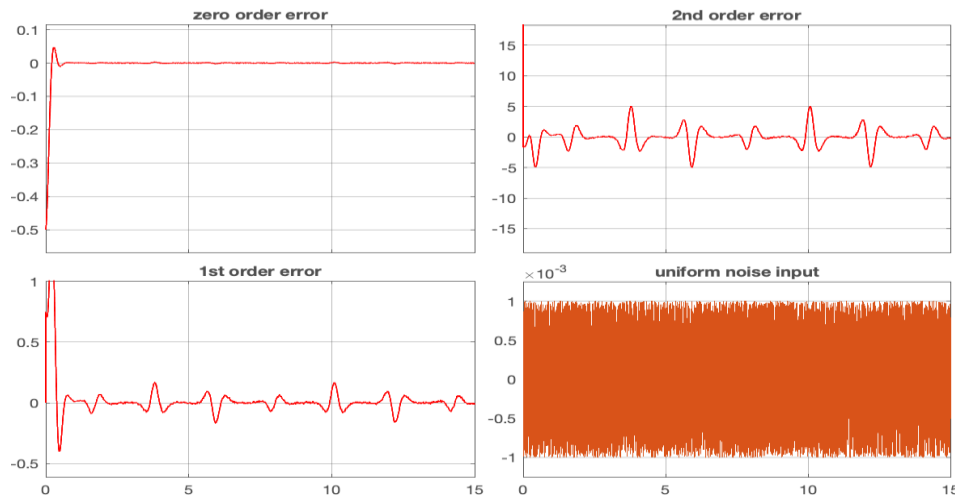


Fig: Error plot of MHGO

HGO estimation with white noise

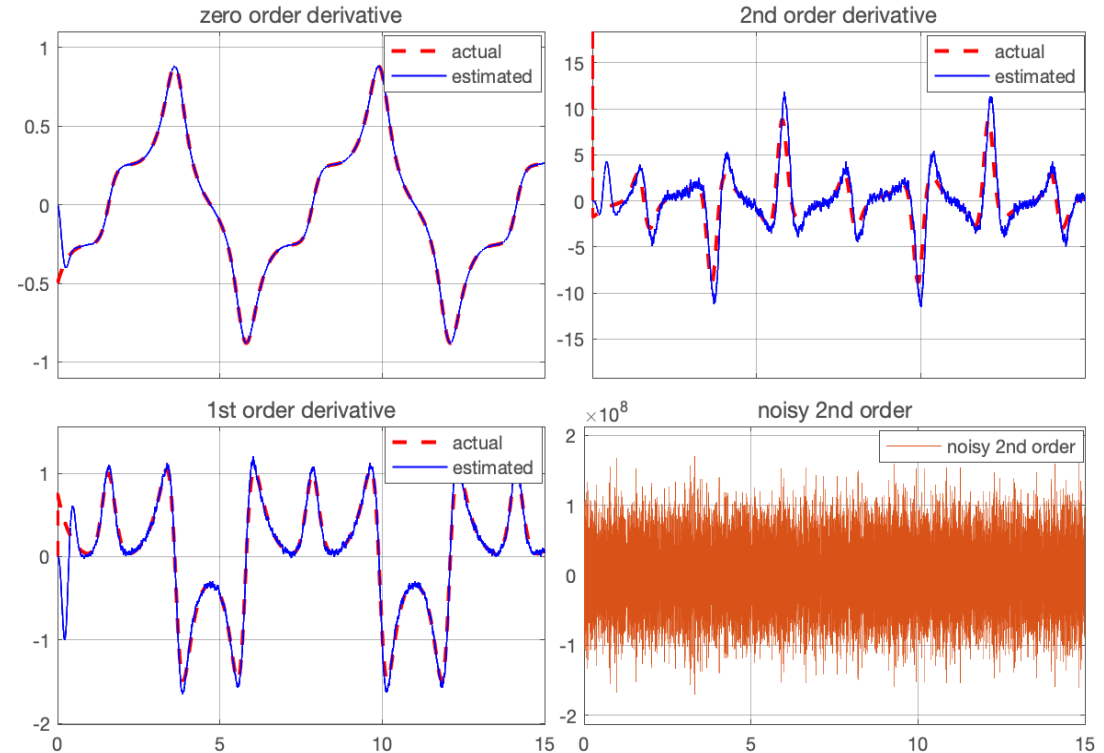


Fig: Actual (red) and estimated (blue) in MHGO

Model-Free Control

Plant description

Airplane roll dynamics

$$\ddot{\beta} = \theta_{10}\beta + \theta_{20}\dot{\beta} + \theta_{60}(\delta_a + d) + (\theta_{30}|\beta| + \theta_{40}|\dot{\beta}|)\dot{\beta} + \theta_{50}\beta^3,$$

It can be rewritten:

$$\ddot{\beta} = \theta_{10}\beta + \theta_{20}\dot{\beta} + \hat{\theta}_6\delta_a + F(\beta, \dot{\beta}, \delta_a, d),$$

$$F(\beta, \dot{\beta}, \delta_a, d) = (\theta_{30}|\beta| + \theta_{40}|\dot{\beta}|)\dot{\beta} + \theta_{50}\beta^3 + \theta_{60}d + (\theta_{60} - \hat{\theta}_6)\delta_a$$

- ✓ This plant is highly uncertain, and noise addition will make it worse

Applied method:

- ✓ Modified high gain observer with extended state – it will estimate the required states and make the system stable.

HGO method for plant dynamics estimation

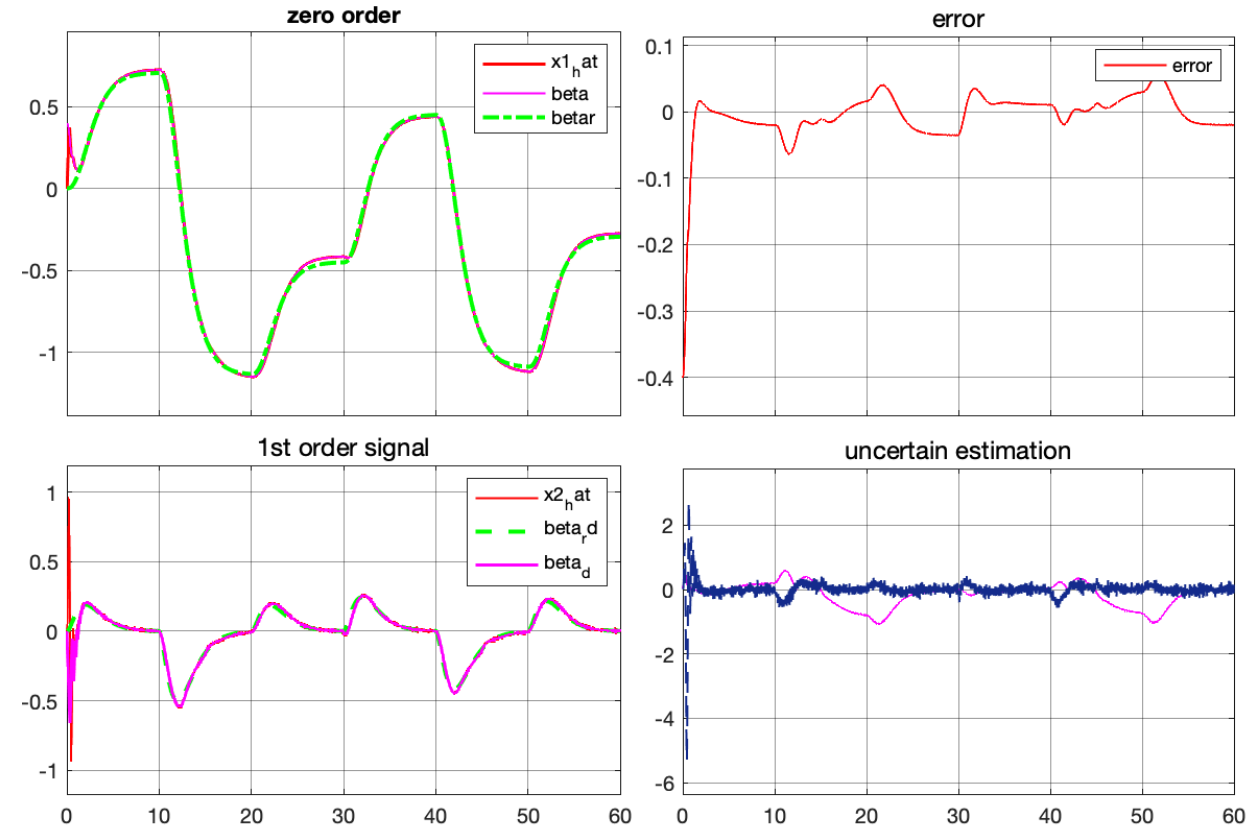


Fig: Simulation result of MFC with MHGO

CONCLUSION AND RECOMMENDATIONS

Conclusion

- The given literatures are reviewed and tested on MATLAB.
- The results helped us to compare which proposed method is more effective. Based on the results, we concluded that HOSM performs better than HGO and HGC due to its noise insensitive and robust nature. Our proposed method also showed better result as well. We have also tested our proposed method on a model-free plant (roll dynamics of a plane) and its result was very good.

Future works

- Test these signal differentiation techniques in IBSC lab to see their effectiveness in real-application
- Review other advanced techniques and compare the result with these methods)

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THANK YOU!
MERCI BEAUCOUP!

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